



VisionFlow

Coordination Engineering for Federated Human–AI Intelligence

Ecosystem Technical Pitch | May 2026

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Federated Substrates

CUDA Kernels

MCP Tools

GPU Speedup

Lines of Rust

OSS Repos

92
185+
55x
120K+
6

EXECUTIVE SUMMARY

VisionFlow is an open-source coordination harness for federated human–AI intelligence. It solves the governance gap that emerges when organisations deploy multiple AI agents: without shared semantics, cryptographic identity, and human-in-the-loop oversight, agents waste 60–80% of tokens on context rediscovery, duplicate reasoning, and contradictory outputs.

The ecosystem comprises six federated repositories, 120K+ lines of Rust, 92 CUDA kernels, and 185+ MCP tools. Five independent substrates mesh through a single cryptographic identity spine (`did:nostr:secp256k1`). Every actor—human, agent, server—is authenticated at every layer via the same keypair, enabling federation without centralised infrastructure.

For token and model vendors, VisionFlow represents the coordination layer that makes extreme token consumption against hard problems rational. Shared ontology eliminates re-derivation. Provenance chains eliminate re-validation. The Judgment Broker eliminates decision spin. The result: 3–5× effective token throughput gain, with every token producing auditable, governed, provenance-tracked reasoning rather than disposable chat.

THE PROBLEM: COORDINATION FAILURE AT SCALE

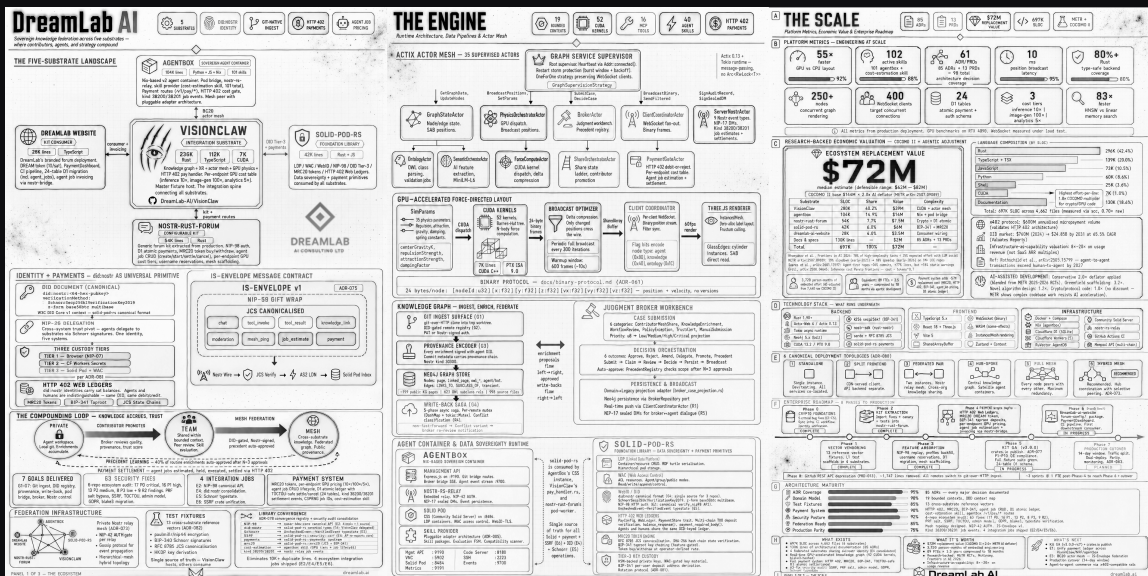
The AI industry has moved through a clear progression: LLM → chatbot → reasoning → agent → agentics → external harness. Each stage solves the previous stage's limitation and reveals a new one. VisionFlow occupies the next position: the **coordination harness**—shared semantics, federated identity, governance, and cryptographic provenance.

73% of frontline AI adoption happens without management sign-off. Your workforce is already building shadow workflows, stitching together AI agents, automating procurement shortcuts. Your organisation is becoming an agentic mesh whether you plan for it or not.

The four failure modes of uncoordinated AI:

- ✗ **Centralised AI** scales tokens but not trust. One vendor, one failure mode, one billing relationship.
- ✗ **Agent frameworks** scale tasks but not governance. Fast and broken is still broken.
- ✗ **Knowledge tools** scale information but not reasoning. More data doesn't mean better decisions.
- ✗ **Collaboration platforms** scale communication but not coordination. More Slack channels don't solve alignment.

ECOSYSTEM ARCHITECTURE: FIVE SUBSTRATES, ONE IDENTITY



Every actor—human, agent, server, worker—shares a single did: `nostr:<hex-pubkey>secp256k1` keypair. This keypair is verified at the Nostr relay, at every HTTP request (NIP-98 Schnorr), against WAC ACLs on Solid pods, in every provenance bead, and is resolvable as a W3C DID Document.

VisionClaw — Knowledge Engineering

- OWL 2 EL formal reasoning via `Whelk-rs` (provably correct entailments, not probabilistic guesses)
- 92 CUDA kernels: semantic relationships become GPU physics forces (`subClassOf` → attraction, `disjointWith` → repulsion), 55× speedup vs CPU
- 934-node force-directed 3D graph with real-time rendering
- 17 MCP ontology tools (7 native + 10 bridge to SPARQL)
- Immersive XR: 15 years of CAVE research at University of Salford, migrating to native Quest 3 APK (Godot 4 + godot-rust + OpenXR)
- THG world record: 250+ concurrent VR users in knowledge graph
- 10 DDD bounded contexts, 114 command/query handlers, CQRS architecture

Maturity: ~120K LOC Rust. Production. GPU physics, graph rendering, ontology tools all operational. XR layer experimental.

Licence: MPL 2.0 (commercial-friendly)

Agentbox — Harness Engineering

- Hardened Nix-built container: byte-for-byte reproducible, non-root, read-only filesystem, capabilities dropped
- 90+ agent skills, 185+ MCP tools (Playwright, ComfyUI, QGIS, Blender, LaTeX, Jupyter, PyTorch)
- Multi-runtime: Claude Code, Codex, Gemini, DeepSeek, Ruflo
- 5-slot pluggable adapter architecture: beads, pods, memory, events, orchestrator
- Embedded Solid Pod server (`solid-pod-rs`) + Nostr relay (`nostr-rs-relay`)
- BIP-340 sovereign identity at bootstrap—agents own their keys
- Privacy filter sidecar: PII redaction before any log or telemetry
- Browser-based setup wizard (zero external dependencies)
- 18 URN kinds for resource naming (`urn:agentbox:<kind>:[<scope>]:<local>`)

Maturity: Production core. 620+ line manifest. 20+ architecture docs (ADRs/PRDs/DDDs). Security audit

hardened May 2026.
Licence: AGPL 3.0

solid-pod-rs — Cryptographic Foundation

- Faithful Rust port of JavaScriptSolidServer (JSS): 98% parity (201 rows tracked, 158 present, 5 partial, 3 missing)
- W3C Solid Protocol: LDP, WAC access control, conditional requests, glob GET
- Dual auth: NIP-98 Schnorr (stateless) + Solid-OIDC with DPOP (RFC 9449)
- Full identity provider crate: OIDC authorization-code flow, dynamic client registration, WebAuthn passkeys, NIP-07 Schnorr SSO
- HTTP 402 micropayments: WAC PaymentCondition, per-read satoshi costs, MRC20 token minting, BIP-341 anchoring
- ActivityPub federation: inbox/outbox, HTTP Signature verification, follower fan-out
- Git HTTP backend: smart protocol, NIP-98 bridge auth, WAC-integrated
- Embedded NIP-01 Nostr relay with DID Document publication
- Dual compile: native Tokio + WASM Cloudflare Workers
- 870+ tests, SSRF hardened, dotfile allowlist, path traversal protection

Maturity: v0.4.0-alpha.15. ~20K LOC across 6 crates.
Feature-complete, pre-1.0 stabilisation.

Licence: AGPL 3.0

nostr-rust-forum — Governance UI & Relay Kit

- 12 Rust crates: core protocol, 5 Cloudflare Workers, Leptos WASM client, mesh federation, rate limiting, config, search, setup
- Judgment Broker decision surface: 1,015-line governance model
- Agent Control Surface Protocol: kinds 31400–31405 (PanelDefinition, PanelState, ActionRequest, ActionResponse, PanelUpdate, PanelRetired)
- REST governance API: 7 NIP-98-gated endpoints (agent registry, broker cases/decisions, roles)
- Passkey-first auth: WebAuthn PRF derives Nostr keys deterministically—no passwords stored
- 3-zone access: public, members, private (runtime configurable)
- Leptos WASM client: 19 pages, 60+ components, governance dashboard, admin panel
- 15 NIP implementations (NIP-01/07/09/29/33/40/42/44/45/50/52/98 + custom 31400–31405)
- Two-tier pod storage: CF Workers (auto-provisioned) + native solid-pod-rs (git-capable)
- Vector search (NIP-50): in-memory cosine k-NN, RVF binary format

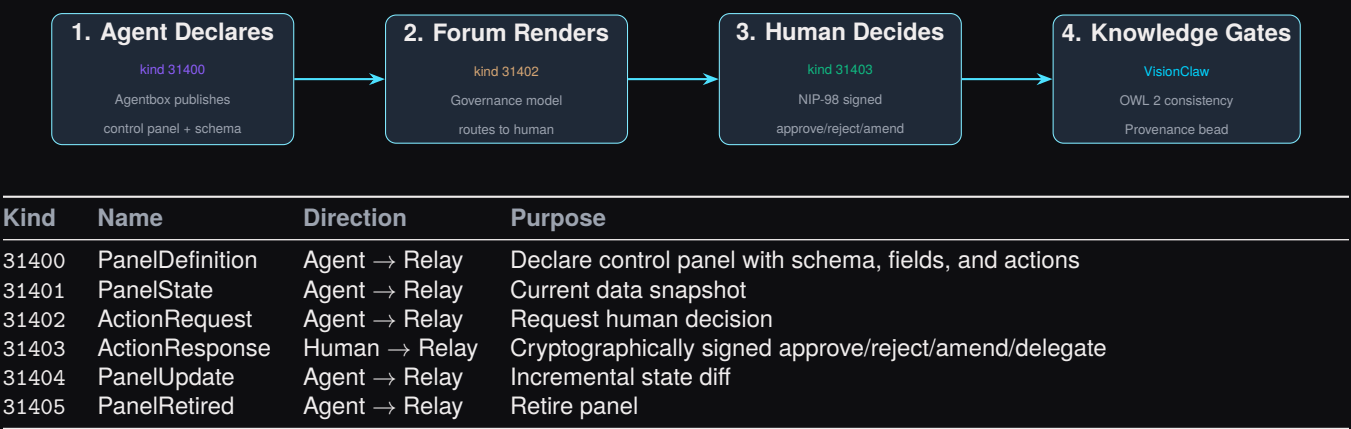
Maturity: v3.0.0-rc7. Phase 1 sprint complete (federated NIP-05, pod-resident keys, data export). Actively deployed.

Licence: MIT / Apache-2.0

 **DreamLab Edge — Branded Deployment** | React SPA + Leptos WASM forum on Cloudflare Workers | Operator overlay configuration | Live at dreamlab-ai.com | Consumes all substrates as library crates

THE JUDGMENT BROKER: HUMAN-IN-THE-LOOP GOVERNANCE THAT ACCELERATES

The Judgment Broker is an emergent property of agents, humans, a knowledge graph, and a relay mesh coordinating through shared cryptographic identity. No single repository owns the broker.



Every decision is an immutable Nostr event. In regulated industries (pharma, finance, defence), the cost of reconstructing decision provenance after the fact dwarfs the cost of generating it by construction. The Judgment Broker provides **90% reduction in audit reconstruction cost**.

TOKEN ECONOMICS: WHY TOKENMAXING FOR HARD PROBLEMS IS RATIONAL

The cost of not coordinating. A 50-person team running uncoordinated AI agents wastes **5–8× tokens** on context rediscovery, duplicate reasoning, and contradictory outputs. Each agent session starts cold—no shared ontology, no provenance, no memory of what other agents concluded.

Coordination as token multiplier. VisionFlow’s shared ontology means agents don’t re-derive domain vocabulary. The provenance chain means agents don’t re-validate conclusions. The Judgment Broker means agents don’t spin on decisions they lack authority to make. Result: **3–5× effective throughput gain**.

Hard problems justify deep spend.

- Drug discovery: \$2.6B average cost per approved compound
- Climate modelling: \$50M+ per actionable simulation suite
- Creative production: \$100M+ per franchise

Against these stakes, \$10K–100K on coordinated AI reasoning is a rounding error—provided the coordination harness prevents even one wrong conclusion from propagating.



The ROI model:

$$ROI = \frac{V \times \text{coordination_mult} \times \text{governance_div}}{\text{token_cost} \times N}$$

The break-even point is typically reached at **3 agents** working on any problem valued above **\$50K**. Beyond that, every additional coordinated agent produces net positive value because the ontology, identity spine, and governance plane are shared infrastructure.

COMPETITIVE LANDSCAPE

Every differentiator sits where no competitor can follow. All competing platforms cluster in the commodity zone.

Platform	Crypto Identity	Data Sovereignty	Governance	Formal Reasoning	Federation	OSS
VisionFlow	✓	✓	✓	✓	✓	✓
Google Spark	✗	✗	~	✗	✗	✗
OpenAI Codex	✗	✗	✗	✗	✗	~
Claude Code	✗	~	~	✗	~	~
Devin	✗	✗	~	✗	✗	✗
CrewAI	✗	~	~	✗	~	✓
AutoGen	✗	~	~	✗	~	✓
LangGraph	✗	~	~	✗	~	✓
Cursor / Windsurf	✗	~	~	✗	✗	✗

✓ Protocol-native ~ Partial/add-on ✗ Absent

Formal Reasoning: Solid Red

Every competitor relies on probabilistic LLM inference. VisionFlow's OWL 2 EL subsumption produces provably correct entailments.

Crypto Identity: Solid Red

Only VisionFlow implements cryptographic agent identity at the protocol level. Every other platform ties identity to someone else's server.

Federation: Solid Red

No competitor supports sovereign nodes trusting each other via cryptographic identity rather than shared infrastructure.

REAL-WORLD SCENARIOS

Climate Modelling Consortium

Three universities, two government agencies, one NGO. Each runs VisionClaw with domain-specific ontologies. OWL 2 EL ensures "sea surface temperature anomaly" means the same thing across all. Data stays in sovereign pods; findings surface through the Judgment Broker.

Pharmaceutical Drug Discovery

Biotech + CRO + regulatory consultancy. Literature mining agents parse 50K papers, chemistry agents run ADMET predictions, compliance agents map to ICH guidelines. Each org's agents operate within their pod boundary. The CRO never sees proprietary target lists.

Creative Production at Scale

12-episode series, five time zones. Production ontology maps episodes to scenes to shots to assets. When a VFX shot depends on an unapproved 3D asset, the OWL 2 constraint propagates through the graph and the GPU physics engine makes the blocked dependency visually obvious.

Real-world validation: DreamLab Creative Hub (50-person team, ~998 graph nodes, daily production) | University of Salford (research partnership, semantic force-directed layout) | THG World Record (250+ concurrent XR users)

SCALING MODEL

👤 Single Operator

One Agentbox, standalone mode. Local SQLite beads, local Solid pod, local events. 90+ skills, 185+ tools, privacy filter active. One API key. One container. Minutes to deploy.

👥 Team (50+)

VisionClaw + forum + Agentbox on a shared relay mesh. Shared ontology, Judgment Broker oversight. Every mutation passes through governance. One GPU host + agent container + CF Workers.

🏢 Federated Enterprise

Multiple sovereign Agentbox instances via Nostr relay mesh. Three transport strata: Tailscale (private WireGuard), Nostr relays (censorship-resistant), Cloudflare Tunnels (edge). Trust via did:nostr, not topology.

Open source. Model-agnostic. Federation-ready.

Every token spent through VisionFlow produces auditable, governed, provenance-tracked reasoning—not disposable chat.

github.com/DreamLab-AI

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